

MODEL QUESTION PAPER
DIPLOMA EXAMINATION IN ENGINEERING /TECHNOLOGY / MANAGEMENT/
COMMERCIAL PRACTISE
BIOMEDICAL INSTRUMENTS

(Maximum Marks:100)

[Time:3Hrs]

PART ---A

(Maximum marks:10)

Marks

I. Answer **all** questions in one or two sentences . Each question carries 2 marks

1. Define diastolic pressure
2. Define bioelectricity
3. List electrodes used for measurement of EEG
4. State the need of ventilator
5. List the scanning methods in Ultra sonic imaging system

(5x2=10)

PART—B

(Maximum marks:30)

II. Answer **any five** of the following questions .Each question carries 6 marks

1. Describe action and resting potential of a cell
2. Explain the principle of piezo electric pulse transducer
3. Describe the block diagram of an EMG machine
4. Explain the chest lead system
5. Explain the functions of hemodialysis machine
6. List the advantages and disadvantages of shortwave diathermy treatment
7. Explain the effect of electricity ,radiation and magnetism in the human body

(5x6=30)

PART—C

Marks

(Maximum marks: 60)

Answer **one full question** from each unit. Each full question carries **15 marks**

Unit -I

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|------|---|---|
| III. | a. Explain the sodium pump and transmission of impulse | 6 |
| | b. Describe the criteria for selecting biomedical transducer | 5 |
| | c. Explain the working principle of strain gauge respiration sensor | 4 |

OR

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| IV. | a. Describe indirect method of blood pressure measurement | 7 |
| | b. Explain working of ultrasonic blood flowmeter with diagram | 8 |

Unit -II

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| V | a. Explain the different electrodes used for ECG measurement | 8 |
| | b. Describe the block diagram of EEG machine | 7 |

OR

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| VI | a. Draw the typical EMG waveform | 3 |
| | b. Explain the block diagram of ECG machine | 8 |
| | c. List the frequency regions of EEG | 4 |

Unit-III

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| VII | a. Classify different types of pacemakers | 7 |
| | b. Explain the working of Ultra sonic diathermy unit | 8 |

OR

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| VIII | a. Describe the working of AC defibrillators | 7 |
| | b. Explain the working of ventricular synchronous demand pacemaker | 8 |

Unit -IV

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| IX | a. Describe the operation of Xray machine | 8 |
| | b. Explain the block diagram of biotelemetry system | 7 |

OR

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|---|--|---|
| X | a. Describe the applications of Biotelemetry | 7 |
| | b. Explain the block diagram of NMR | 8 |

(15x4=60)